

THE SMASH-POWER SEGAL CONJECTURE WITHOUT FINITE-TYPE HYPOTHESES

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ABSTRACT. Let $G \neq 1$ be a finite p -group and let B be a bounded-below spectrum. We prove that the proper Tate diagonal

$$\Delta_G(B): B \longrightarrow (B^{\wedge G})^{\tau G}$$

is the stable p -completion map. Equivalently, the comparison from the categorical fixed points of the Hill–Hopkins–Ravenel norm $N_e^G B$ to its homotopy fixed points is a mod- p equivalence. The published theorem of Bergsaker–Rognes proves this under a degreewise finite mod- p homology hypothesis. The new argument uses their theorem only for finite-type Eilenberg–Mac Lane input. The finite-type hypothesis is removed by studying the representation-sphere obstruction

$$Q_G(B) = F_G(S^{\infty \bar{\rho} G}, N_e^G B)^G.$$

Finite representation stages, a uniform pro-zero argument, exactness, and a Postnikov-continuity statement specific to norm inputs imply that $Q_G(B)/p \simeq 0$ for every bounded-below B . Isotropy separation and induction on $|G|$ then finish the argument. An appendix records why the stronger assertion that proper Tate constructions commute with arbitrary bounded-below Postnikov towers is false.

Status note. This manuscript records a proof assembled from published equivariant inputs and several new formal lemmas. It has not undergone external peer review.

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1. INTRODUCTION

For a finite group G , let $N_e^G: \mathrm{Sp} \rightarrow \mathrm{Sp}^G$ denote the Hill–Hopkins–Ravenel norm. Its underlying Borel G -spectrum is the G -fold smash power $B^{\wedge G}$, with G acting by the left regular permutation of the factors. The fixed-to-homotopy-fixed comparison is

$$\gamma_{G,B}: (N_e^G B)^G \longrightarrow (B^{\wedge G})^{hG}.$$

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Bergsaker and Rognes proved that $\gamma_{G,B}$ is a mod- p equivalence when G is a finite p -group, B is bounded below, and $H_*(B; \mathbf{F}_p)$ is degreewise finite [2, Theorem 1.1]. Nikolaus conjectured that the finite-type assumption is unnecessary [5, Conjecture 7.10 and Remark 7.11].

The purpose of this paper is to prove the following statement.

Theorem 1.1 (Smash-power Segal conjecture). *Let $G \neq 1$ be a finite p -group and let B be a bounded-below spectrum. Then*

$$\gamma_{G,B}: (N_e^G B)^G \longrightarrow (B^{\wedge G})^{hG}$$

is a mod- p equivalence.

Our convention is that stable p -completion means Bousfield localization at the mod- p Moore spectrum:

$$X_p^\wedge = L_{S/p} X.$$

Thus a map is a mod- p equivalence precisely when its cofiber becomes contractible after smashing with S/p .

The proper Tate construction is denoted by $(-)^{\tau G}$; this is also written $(-)^{\varphi G}$ in [5, 8]. Applying geometric fixed points to the Borelification unit for $N_e^G B$ gives the proper Tate diagonal

$$\Delta_G(B): B \simeq \Phi^G N_e^G B \longrightarrow \Phi^G \beta_G(B^{\wedge G}) \simeq (B^{\wedge G})^{\tau G};$$

see [8, Example 2.28]. Quigley–Shah prove that the target is p -complete for bounded-below input [7, Lemma 3.15]. We will therefore deduce the following form of Nikolaus’s conjecture.

Theorem 1.2 (Proper Tate diagonal). *Let $G \neq 1$ be a finite p -group and let B be a bounded-below spectrum. The map*

$$\Delta_G(B): B \longrightarrow (B^{\wedge G})^{\tau G}$$

exhibits its target as $B_p^\wedge$.

The proof does *not* assert that proper Tate constructions commute with Postnikov limits on arbitrary Borel G -spectra. That assertion is false; see [section A](#). Instead we isolate the norm-specific obstruction

$$Q_G(B) = F_G(S^{\infty \bar{\rho}_G}, N_e^G B)^G,$$

prove the particular continuity property needed for Q_G , and show that $Q_G(B)/p \simeq 0$.

Here is the proof strategy. The tensor-cube filtration first shows that the obstruction functor Q_G is exact: every mixed tensor word is induced from a proper subgroup and is killed by the representation-sphere tower. For $H = H\mathbf{F}_p$, the finite representation stages have finite p -primary homotopy, while the published finite-type theorem implies $Q_G(H) \simeq 0$. Consequently the stage towers are uniformly pro-zero, which allows us to pass from one copy of H to an arbitrary wedge HV , even when V is infinite-dimensional. Exactness then handles p^2 -torsion Eilenberg–Mac Lane layers. Finally, the formula for geometric fixed points of a regular norm supplies a *norm-specific* connectivity estimate and hence Postnikov continuity:

$$\begin{aligned} \text{finite stages} &\implies \text{uniform pro-zero} \implies \text{arbitrary } V, \\ \text{norm connectivity} &\implies \text{Postnikov dévissage.} \end{aligned}$$

This gives $Q_G(B)/p \simeq 0$. Proper-family isotropy separation and induction on $|G|$ then prove the Segal comparison, and Borelification identifies its geometric-fixed-point image with the proper Tate diagonal.

2. EQUIVARIANT CONVENTIONS AND STANDARD INPUTS

Write Sp^G for genuine G -spectra and $\mathrm{Sp}^{hG} = \mathrm{Fun}(BG, \mathrm{Sp})$ for Borel G -spectra. Let $\mathcal{P} = \mathcal{P}_G$ be the family of proper subgroups of G . We use $E\mathcal{P}$ for its universal G -space and $\tilde{E}\mathcal{P}$ for the cofiber of $E\mathcal{P}_+ \rightarrow S^0$.

Let $\bar{\rho}_G$ be the reduced *complex* regular representation of G . For each $K \leq G$,

$$\dim_{\mathbf{C}}(\bar{\rho}_G^K) = [G : K] - 1, \quad \dim_{\mathbf{R}}(\bar{\rho}_G^K) = 2([G : K] - 1).$$

The factor 2 in later connectivity estimates comes from this use of the complex, rather than real, regular representation. It follows that

$$S^{\infty \bar{\rho}_G} := \operatorname{colim}_m S^{m \bar{\rho}_G} \simeq \tilde{E}\mathcal{P}.$$

Indeed, its K -fixed points are contractible for $K < G$ and equal to S^0 for $K = G$.

We distinguish three finiteness conditions. A spectrum has *degreewise finite mod- p homology* if every $H_i(-; \mathbf{F}_p)$ is finite-dimensional. It has *degreewise finitely generated homotopy* if every integral homotopy group is finitely generated, and it has *degreewise finite homotopy* if every homotopy group is a finite abelian group. These conditions will not be conflated under the single phrase “finite type.”

We use the norm restriction and geometric-fixed-point formulas

$$\operatorname{Res}_H^G N_e^G B \simeq N_e^H(B^{\wedge H \setminus G}), \quad (1)$$

$$\Phi^H N_e^G B \simeq B^{\wedge H \setminus G}. \quad (2)$$

These are standard properties of the norm; see [3, Appendix B] and [2, Proposition 2.1].

We will repeatedly use the following fixed/geometric-fixed comparison.

Lemma 2.1 (Fixed-point detection of connectivity). *Let $X \in \mathrm{Sp}^G$.*

- (1) *If $\Phi^K X$ is c -connective for every $K \leq G$, then X^K is c -connective for every $K \leq G$.*
- (2) *The fixed-point spectra X^K are bounded below and have degreewise finitely generated homotopy for every $K \leq G$ if and only if the same is true of all $\Phi^K X$.*

Proof. We prove the first assertion directly by induction on the order of a subgroup $H \leq G$. For $H = e$, categorical and geometric fixed points agree. Suppose the assertion is known for every proper subgroup of H , and let $\mathcal{P}_H$ denote the family of proper subgroups of H . Isotropy separation for $\operatorname{Res}_H^G X$, followed by H -fixed points, gives a cofiber sequence

$$(E\mathcal{P}_{H+} \wedge \operatorname{Res}_H^G X)^H \longrightarrow X^H \longrightarrow (\tilde{E}\mathcal{P}_H \wedge \operatorname{Res}_H^G X)^H = \Phi^H X.$$

The first term is c -connective. Indeed, build $E\mathcal{P}_H$ skeletally. An equivariant cell with isotropy $L < H$, after smashing with $\operatorname{Res}_H^G X$ and taking H -fixed points, is a nonnegative suspension of X^L , which is c -connective by induction. Connective spectra are closed under extensions and filtered colimits, so the full cellular object is c -connective. The last term is c -connective by hypothesis; hence so is X^H . This completes the induction.

The second assertion, with “finite type” interpreted as degreewise finitely generated homotopy, is [2, Lemma 6.3]. Its proof is the same finite isotropy induction, now applied simultaneously to boundedness and finite generation. $\square$

We also use two standard facts without further mention. First, the geometric fixed-point functors $\{\Phi^K\}_{K \leq G}$ are jointly conservative. Second, categorical fixed points by a finite group preserve filtered colimits: the orbit spectra G/K_+ , $K \leq G$, are compact. In particular, categorical fixed points commute with smashing by $S_{(p)}$ and by $H\mathbf{Q}$, since these localizations are filtered colimits of finite spectra.

3. THE REPRESENTATION-SPHERE OBSTRUCTION

Definition 3.1. For a genuine G -spectrum Y , set

$$\begin{aligned}\mathcal{Q}_{G,m}(Y) &= F_G(S^{m\bar{\rho}_G}, Y)^G, \\ \mathcal{Q}_G(Y) &= \operatorname{holim}_m \mathcal{Q}_{G,m}(Y) \simeq F_G(S^{\infty\bar{\rho}_G}, Y)^G.\end{aligned}$$

For an ordinary spectrum B , define

$$Q_{G,m}(B) = \mathcal{Q}_{G,m}(N_e^G B), \quad Q_G(B) = \mathcal{Q}_G(N_e^G B).$$

The inclusions $S^{m\bar{\rho}_G} \rightarrow S^{(m+1)\bar{\rho}_G}$ define all the inverse towers above.

The spectrum $Q_G(B)$ is the obstruction appearing in proper-family isotropy separation. We first prove a vanishing statement for induced towers. The use of Wirthmüller ambidexterity in this lemma is essential.

Lemma 3.2 (Induced Euler towers). *Let $K < G$, let $A \in \operatorname{Sp}^G$, and let $Z \in \operatorname{Sp}^K$. The transition maps in the tower*

$$T_m = F_G(S^{m\bar{\rho}_G}, A \wedge \operatorname{Ind}_K^G Z)^G$$

are nullhomotopic. Consequently $\operatorname{holim}_m T_m \simeq 0$.

Proof. The projection formula gives

$$A \wedge \operatorname{Ind}_K^G Z \simeq \operatorname{Ind}_K^G (\operatorname{Res}_K^G A \wedge Z).$$

For a finite group, induction and coinduction agree. Wirthmüller ambidexterity therefore identifies

$$T_m \simeq F_K(\operatorname{Res}_K^G S^{m\bar{\rho}_G}, \operatorname{Res}_K^G A \wedge Z)^K.$$

The transition $T_{m+1} \rightarrow T_m$ is precomposition with the restriction of the Euler map

$$a_{\bar{\rho}_G}: S^0 \longrightarrow S^{\bar{\rho}_G}.$$

Since $K < G$, the restricted representation has a positive-dimensional trivial summand:

$$\operatorname{Res}_K^G \bar{\rho}_G \cong (\bar{\rho}_G)^K \oplus W, \quad \dim_{\mathbf{C}}(\bar{\rho}_G)^K = [G : K] - 1 > 0.$$

Under this decomposition, the Euler map factors as the smash product of the Euler maps of $(\bar{\rho}_G)^K$ and W . The first factor is a stable map $S^0 \rightarrow S^{2([G:K]-1)}$ with trivial K -action and is null because the sphere spectrum has no negative homotopy groups. Thus $\operatorname{Res}_K^G(a_{\bar{\rho}_G})$ is K -equivariantly nullhomotopic, and every transition map is nullhomotopic.

On homotopy groups the tower therefore has zero transition homomorphisms. Both $\lim$ and $\lim^1$ of such a tower vanish, so the Milnor exact sequence gives $\pi_* \operatorname{holim}_m T_m = 0$. Hence the homotopy limit is contractible. $\square$

Lemma 3.3 (Equivariant tensor-cube filtration). *Let*

$$B' \longrightarrow B \longrightarrow B''$$

be a cofiber sequence of spectra and put $n = |G|$. There is a natural finite G -equivariant filtration

$$F_0 \longrightarrow F_1 \longrightarrow \cdots \longrightarrow F_n = N_e^G B$$

with $F_0 = N_e^G B'$ and

$$\operatorname{gr}_j F := \operatorname{cofib}(F_{j-1} \rightarrow F_j) \simeq \bigvee_{\substack{S \subseteq G \\ |S|=j}} (B')^{\wedge(G \setminus S)} \wedge (B'')^{\wedge S}. \quad (3.1)$$

In particular, $\mathrm{gr}_n F \simeq N_e^G B''$. If $0 < j < n$, then each orbit summand in (3.1) is of the form

$$\mathrm{Ind}_{G_S}^G W_S, \quad G_S = \{g \in G \mid gS = S\} < G,$$

where $W_S = (B')^{\wedge(G \setminus S)} \wedge (B'')^{\wedge S}$ carries its canonical G_S -action.

Proof. Form the G -cube obtained by smashing, in the positions indexed by G , copies of the map $B' \rightarrow B$. Filter its terminal vertex by the number of coordinates in which the cofiber B'' is used. The standard cubical filtration in a stable symmetric monoidal category gives the stated finite filtration and the graded formula (3.1). The construction is natural in the original cofiber sequence.

The left regular action of G sends the word indexed by S to the word indexed by gS . For a chosen S , the wedge over its orbit is therefore canonically

$$\bigvee_{S' \in G \cdot S} W_{S'} \simeq G_+ \wedge_{G_S} W_S = \mathrm{Ind}_{G_S}^G W_S.$$

If $G_S = G$, then S is invariant under every left translation. Since G acts transitively on itself, this forces $S = \emptyset$ or $S = G$. Thus $G_S < G$ for every nonempty proper S . $\square$

Proposition 3.4 (Exactness). *The functor $\mathcal{Q}_G: \mathrm{Sp} \rightarrow \mathrm{Sp}$ is exact.*

Proof. The functor $\mathcal{Q}_G: \mathrm{Sp}^G \rightarrow \mathrm{Sp}$ is exact, being a limit of the exact functors $F_G(S^{m\bar{p}_G}, -)^G$. By lemma 3.2, it annihilates spectra induced from proper subgroups.

Apply $\mathcal{Q}_G$ to the finite filtration in lemma 3.3. Exactness of $\mathcal{Q}_G$ and the induced-orbit description show that all graded pieces with $0 < j < |G|$ vanish. Hence $\mathcal{Q}_G(F_0) \rightarrow \mathcal{Q}_G(F_{|G|-1})$ is an equivalence, and the last filtration cofiber sequence becomes

$$\mathcal{Q}_G(F_0) \longrightarrow \mathcal{Q}_G(F_{|G|}) \longrightarrow \mathcal{Q}_G(N_e^G B'').$$

Using $F_0 = N_e^G B'$ and $F_{|G|} = N_e^G B$, this is

$$Q_G(B') \longrightarrow Q_G(B) \longrightarrow Q_G(B'').$$

The first arrow is induced by $N_e^G(B' \rightarrow B)$. Naturality of the cubical filtration identifies the second arrow with $\mathcal{Q}_G$ applied to $N_e^G(B \rightarrow B'')$: on the top graded quotient it is the identity of $N_e^G B''$, while it vanishes on all earlier filtration pieces. Thus the arrows are exactly those obtained by applying Q_G to the original cofiber sequence, not merely abstract arrows between equivalent objects. $\square$

Lemma 3.5 (Exact functors are additive). *Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor between stable ∞ -categories. Then F preserves finite biproducts and, for every integer r ,*

$$F(r \cdot \mathrm{id}_X) = r \cdot \mathrm{id}_{F(X)}.$$

Proof. An exact functor preserves the zero object and finite coproducts, since these are finite pushouts. In a stable category finite coproducts are also finite products, so F preserves biproducts. For $r \geq 0$, the map $r \cdot \mathrm{id}_X$ is the composite of the diagonal $X \rightarrow X^{\oplus r}$, the direct sum of r identity maps, and the fold $X^{\oplus r} \rightarrow X$. Applying F gives the same construction on $F(X)$. Additive inverses in a stable category are preserved as well, which handles negative r . $\square$

4. THE FINITE-TYPE EILENBERG–MAC LANE CORE

Put $H = H\mathbf{F}_p$. We begin with the finite representation stages.

Lemma 4.1 (Finite stages). *For every $m \geq 0$, the spectrum $Q_{G,m}(H)$ is bounded below, and each group $\pi_q Q_{G,m}(H)$ is a finite p -group.*

Proof. Let

$$Z_m = S^{-m\bar{\rho}_G} \wedge N_e^G H \in \mathrm{Sp}^G.$$

By (2),

$$\Phi^K Z_m \simeq \Sigma^{-2m([G:K]-1)} H^{\wedge K \setminus G}. \quad (4.1)$$

Put $r = |K \setminus G|$. Iterated Künneth isomorphisms over $\mathbf{F}_p$ identify, as graded vector spaces,

$$\pi_*(H^{\wedge r}) \cong \mathcal{A}_*^{\otimes_{\mathbf{F}_p}(r-1)},$$

where $\mathcal{A}_* = \pi_*(H \wedge H)$ is the dual Steenrod algebra. The graded algebra $\mathcal{A}_*$ is connective and finite-dimensional in each degree. Since r is finite, the same is true of its displayed tensor power. Thus every spectrum in (4.1) is bounded below and has degreewise finite homotopy. The finite-generation part of lemma 2.1 now implies that $Z_m^G = Q_{G,m}(H)$ is bounded below with degreewise finitely generated integral homotopy groups.

It remains to identify their primary support. Consider the localization map

$$Z_m \longrightarrow Z_m \wedge S_{(p)}.$$

After applying any Φ^K , this is an equivalence by (4.1), because an iterated smash power of H is p -local. Joint conservativity of all geometric fixed-point functors therefore shows that the localization map is a genuine equivalence. Likewise all geometric fixed points of $Z_m \wedge H\mathbf{Q}$ vanish, so joint conservativity gives $Z_m \wedge H\mathbf{Q} \simeq 0$.

Categorical fixed points commute with these two localizations: both $S_{(p)}$ and $H\mathbf{Q}$ are filtered colimits of finite spectra, and the orbit representing categorical fixed points is compact. Hence $Q_{G,m}(H) = Z_m^G$ is p -local and rationally contractible. Each of its homotopy groups is therefore a finitely generated, p -local torsion abelian group, hence a finite p -group. $\square$

Corollary 4.2. *The spectrum $Q_G(H)$ is p -complete.*

Proof. A bounded-below spectrum with finite p -group homotopy groups is p -complete: apply Postnikov completeness and note that p -complete spectra are closed under limits. Thus every $Q_{G,m}(H)$ is p -complete by lemma 4.1. Their inverse limit $Q_G(H)$ is p -complete as well. $\square$

We next combine the finite-type theorem of Bergsaker–Rognes with proper isotropy separation. Let $Y = N_e^G H$. There is a commutative square

$$\begin{array}{ccc} Y^G & \xrightarrow{\gamma_{G,H}} & F_G(EG_+, Y)^G \\ \lambda_{G,H} \downarrow & & \simeq \downarrow r_{G,H} \\ F_G(E\mathcal{P}_+, Y)^G & \xrightarrow{\gamma_{\mathcal{P},H}} & F_G(EG_+ \wedge E\mathcal{P}_+, Y)^G. \end{array} \quad (3)$$

Here $\gamma_{G,H}$ is the fixed-to-homotopy-fixed comparison, $\lambda_{G,H}$ is induced by $E\mathcal{P}_+ \rightarrow S^0$, $\gamma_{\mathcal{P},H}$ is the comparison after mapping out of $E\mathcal{P}_+$, and $r_{G,H}$ is induced by the projection $EG \times E\mathcal{P} \rightarrow EG$. The fiber of $\lambda_{G,H}$ is $Q_G(H)$.

Proposition 4.3 (Vanishing of the core obstruction). *For every nontrivial finite p -group G ,*

$$Q_G(H) \simeq 0.$$

Proof. The top horizontal map in (3) is a mod- p equivalence by [2, Theorem 1.1]. For each proper $K < G$, the cellwise comparison entering the lower horizontal map is

$$Y^K \longrightarrow (\mathrm{Res}_K^G Y)^{hK}.$$

By (1),

$$\mathrm{Res}_K^G Y \simeq N_e^K (H^{\wedge K \setminus G}).$$

The coefficient $H^{\wedge K \setminus G}$ is bounded below with degreewise finite mod- p homology, so the same theorem of Bergsaker–Rognes applies. Hence every cellwise comparison is a mod- p equivalence. Since S/p is dualizable, smashing with S/p commutes with the homotopy limits obtained by mapping out of the G -CW complex $E\mathcal{P}$. Thus the lower horizontal map is a mod- p equivalence.

The right vertical map is an equivalence because $EG \times E\mathcal{P}$ is a free contractible G -space. It follows that the left vertical map is a mod- p equivalence, so its fiber $Q_G(H)$ is mod- p acyclic. By [corollary 4.2](#), it is also p -complete. A spectrum that is both S/p -local and S/p -acyclic is contractible. $\square$

5. UNIFORM PRO-ZERO TOWERS AND ARBITRARY VECTOR SPACES

The finiteness of the stages in [lemma 4.1](#) permits arbitrary direct sums because every summand uses the identical tower.

Lemma 5.1 (Uniform pro-zero). *Let*

$$\cdots \longrightarrow A_2 \longrightarrow A_1 \longrightarrow A_0$$

be an inverse sequence of finite abelian groups. If $\lim_m A_m = 0$, then the tower is pro-zero. Moreover, for every set I , the tower $\{\bigoplus_I A_m\}_m$ is pro-zero.

Proof. Fix n . The subgroups

$$\mathrm{im}(A_m \rightarrow A_n), \quad m \geq n,$$

form a descending sequence in the finite group A_n , hence stabilize. The stable image is the image of $\lim_m A_m \rightarrow A_n$. To see this, an element in every image admits a compatible family of lifts by compactness of inverse limits of nonempty finite sets. The stable image is therefore zero. This is precisely the pro-zero condition.

For $\bigoplus_I A_m$, the same index $m = m(n)$ that kills $A_m \rightarrow A_n$ kills every summand simultaneously. Thus it also kills $\bigoplus_I A_m \rightarrow \bigoplus_I A_n$. $\square$

Lemma 5.2 (Spectrum form). *Let $\{E_m\}_m$ be a tower of spectra such that every $\pi_q E_m$ is finite and $\mathrm{holim}_m E_m \simeq 0$. Then, for every set I ,*

$$\mathrm{holim}_m \bigvee_{i \in I} E_m \simeq 0.$$

Proof. For each q , the Milnor exact sequence and finiteness show that $\{\pi_q E_m\}_m$ has zero inverse limit. By [lemma 5.1](#), the tower $\{\bigoplus_I \pi_q E_m\}_m$ is pro-zero, so its inverse limit and $\lim^1$ vanish. The Milnor exact sequence for $\mathrm{holim}_m \bigvee_I E_m$ gives the result. $\square$

Proposition 5.3 (Arbitrary $\mathbf{F}_p$ -vector spaces). *For every $\mathbf{F}_p$ -vector space V , with no dimension restriction,*

$$Q_G(HV) \simeq 0.$$

Proof. Choose a basis $V = \bigoplus_{i \in I} \mathbf{F}_p$, and write H_i for the copy of H indexed by i . Then

$$HV \simeq \bigvee_{i \in I} H_i.$$

For a function $f: G \rightarrow I$, define its tensor word by

$$Z_f = \bigwedge_{x \in G} H_{f(x)}.$$

Distributivity of the indexed smash product gives a genuine G -equivariant decomposition

$$N_e^G(HV) \simeq \bigvee_{f: G \rightarrow I} Z_f. \quad (5.1)$$

The action on functions is

$$(a \cdot f)(x) = f(a^{-1}x),$$

and the symmetry of the smash product supplies compatible maps $Z_f \rightarrow Z_{a \cdot f}$. Thus the stabilizer

$$G_f = \{a \in G \mid f(a^{-1}x) = f(x) \text{ for all } x \in G\}$$

acts naturally on Z_f , and for every word orbit there is a canonical spectrum-level equivalence

$$\bigvee_{f' \in G \cdot f} Z_{f'} \simeq G_+ \wedge_{G_f} Z_f = \text{Ind}_{G_f}^G Z_f. \quad (5.2)$$

Because the left regular G -set G is transitive, $G_f = G$ exactly when f is constant.

For fixed m , the functor

$$\mathcal{Q}_{G,m}(Y) = F_G(S^{m\bar{\rho}_G}, Y)^G$$

commutes with arbitrary wedges. One way to see this is that the composite is the mapping spectrum represented by the finite genuine G -cell $G/G_+ \wedge S^{m\bar{\rho}_G}$, which is compact. Equivalently, $S^{m\bar{\rho}_G}$ is dualizable, and categorical fixed points preserve finite wedges and filtered colimits. Consequently (5.1) and (5.2) split the m -th stage as the wedge of a constant-word part and a nonconstant-word part.

This splitting is a splitting of inverse *systems*. Indeed, the transition $\mathcal{Q}_{G,m+1}(Y) \rightarrow \mathcal{Q}_{G,m}(Y)$ is precomposition with the Euler map of $\bar{\rho}_G$ and is the identity on Y ; hence it preserves every word orbit. The constant words are indexed by I , and their subsystem is precisely

$$\left\{ \bigvee_{i \in I} \mathcal{Q}_{G,m}(H) \right\}_m.$$

By lemmas 4.1 and 5.2 and proposition 4.3, its homotopy limit is contractible.

For the nonconstant words, group the orbit summands according to their stabilizer. Since G has only finitely many subgroups, this rewrites the nonconstant subsystem as a finite wedge of towers of the form

$$\left\{ F_G \left(S^{m\bar{\rho}_G}, \text{Ind}_K^G \left(\bigvee_{[f]: G_f = K} Z_f \right) \right)^G \right\}_m, \quad K < G.$$

Here the inner wedge may be infinite, but it is a single genuine K -spectrum. By lemma 3.2, every transition in each such tower is nullhomotopic and its homotopy limit is contractible. The constant and nonconstant inverse subsystems therefore both have contractible homotopy limit, proving $Q_G(HV) \simeq 0$. $\square$

6. NORM-SPECIFIC POSTNIKOV CONTINUITY

We use the following convention throughout this section: a spectrum X is c -connective if $\pi_i X = 0$ for $i < c$, and a map is c -connective if its homotopy fiber is c -connective.

Lemma 6.1 (Connectivity of a finite smash power). *Let $f: X \rightarrow Y$ be a map between d -connective spectra, and suppose that $\text{fib}(f)$ is c -connective with $c \geq d$. For every $r \geq 1$, the fiber of*

$$f^{\wedge r}: X^{\wedge r} \longrightarrow Y^{\wedge r}$$

is $c + (r - 1)d$ -connective.

Proof. Factor $f^{\wedge r}$ by replacing the r factors one at a time. The fiber of each successive map is a smash product of $\text{fib}(f)$ with $r - 1$ factors, each equal to either X or Y . It is therefore $c + (r - 1)d$ -connective. The fiber of the composite has a finite filtration whose graded pieces are these successive fibers. The full subcategory of $c + (r - 1)d$ -connective spectra is extension-closed, which proves the claim. $\square$

We now prove the continuity statement that replaces the false general continuity of proper Tate constructions.

Proposition 6.2 (Continuity of Q_G). *If B is bounded below, then the natural map*

$$Q_G(B) \longrightarrow \text{holim}_n Q_G(\tau_{\leq n} B)$$

is an equivalence.

Proof. Choose $d \in \mathbf{Z}$ such that B is d -connective, and put $B_n = \tau_{\leq n} B$. The fiber of $B \rightarrow B_n$ is $\tau_{\geq n+1} B$, hence is $(n + 1)$ -connective. Fix a representation stage m . For a subgroup $K \leq G$, put $r_K = |K \backslash G|$. The geometric-fixed-point formula gives a map

$$\Sigma^{-2m([G:K]-1)} B^{\wedge r_K} \longrightarrow \Sigma^{-2m([G:K]-1)} B_n^{\wedge r_K}. \quad (6.1)$$

For $n + 1 \geq d$, [lemma 6.1](#) shows that its fiber is at least

$$c_{m,K}(n) = n + 1 + (r_K - 1)d - 2m([G:K] - 1) \quad (6.2)$$

connective. For fixed m and K , this tends to $+\infty$ with n . Because G has only finitely many subgroups, the minimum

$$c_m(n) = \min_{K \leq G} c_{m,K}(n)$$

also tends to $+\infty$.

Let $F_{m,n}$ be the fiber in Sp^G of

$$S^{-m\bar{\rho}_G} \wedge N_e^G B \longrightarrow S^{-m\bar{\rho}_G} \wedge N_e^G B_n.$$

For every $K \leq G$, formula (6.1) identifies $\Phi^K F_{m,n}$ with a $c_m(n)$ -connective spectrum. Applying [lemma 2.1](#) to $F_{m,n}$ therefore shows that $F_{m,n}^G$ is $c_m(n)$ -connective. Equivalently, the fiber of

$$Q_{G,m}(B) \longrightarrow Q_{G,m}(B_n)$$

has connectivity tending to $+\infty$. The fiber of the induced map to the homotopy limit is the homotopy limit of these fibers and is contractible. Thus, for every fixed m ,

$$Q_{G,m}(B) \simeq \text{holim}_n Q_{G,m}(B_n).$$

Finally, homotopy limits are small limits in spectra, and small limits commute with one another. Hence

$$\begin{aligned} Q_G(B) &\simeq \text{holim}_m \text{holim}_n Q_{G,m}(B_n) \\ &\simeq \text{holim}_n \text{holim}_m Q_{G,m}(B_n) \\ &= \text{holim}_n Q_G(B_n), \end{aligned}$$

as required. $\square$

Remark 6.3. [Proposition 6.2](#) uses the special geometric-fixed-point formula for regular norm inputs. It does not imply that

$$E^{\tau G} \longrightarrow \text{holim}_n (\tau_{\leq n} E)^{\tau G}$$

is an equivalence for arbitrary Borel G -spectra E . The latter is false by [section A](#).

7. POSTNIKOV DÉVISSAGE

Lemma 7.1. *If an abelian group M satisfies $p^2M = 0$, then*

$$Q_G(HM) \simeq 0.$$

Proof. There is a short exact sequence

$$0 \longrightarrow pM \longrightarrow M \longrightarrow M/pM \longrightarrow 0.$$

Both pM and M/pM are $\mathbf{F}_p$ -vector spaces. The associated Eilenberg–Mac Lane spectra form a cofiber sequence. Apply the exact functor Q_G , use [proposition 5.3](#) on the two endpoints, and conclude by two-out-of-three. $\square$

Proposition 7.2 (Vanishing modulo p). *For every bounded-below spectrum B ,*

$$Q_G(B)/p \simeq 0.$$

Proof. Let $B/p = \text{cofib}(p: B \rightarrow B)$. Its long exact homotopy sequence contains

$$0 \longrightarrow \pi_j(B)/p \longrightarrow \pi_j(B/p) \longrightarrow \pi_{j-1}(B)[p] \longrightarrow 0.$$

Consequently $p^2\pi_j(B/p) = 0$ for every j .

Each finite Postnikov section $\tau_{\leq n}(B/p)$ has a finite filtration with layers $\Sigma^j H\pi_j(B/p)$. By [lemma 7.1](#) and [proposition 3.4](#),

$$Q_G(\tau_{\leq n}(B/p)) \simeq 0.$$

The continuity of [proposition 6.2](#) yields

$$Q_G(B/p) \simeq 0.$$

Apply the exact functor Q_G to the cofiber sequence

$$B \xrightarrow{p} B \longrightarrow B/p.$$

The point here is that the *composite* functor Q_G is exact by [proposition 3.4](#); the norm functor by itself is not additive. By [lemma 3.5](#), exactness forces

$$Q_G(p \cdot \text{id}_B) = p \cdot \text{id}_{Q_G(B)}.$$

Since the cofiber $Q_G(B/p)$ is contractible, multiplication by p is an equivalence on $Q_G(B)$. Equivalently, $Q_G(B)/p \simeq 0$. $\square$

8. ISOTROPY SEPARATION AND THE SEGAL COMPARISON

Proof of theorem 1.1. We argue by induction on $|G|$. The assertion for the trivial group is tautological and serves only as the base of the induction. Let $G \neq 1$ and suppose the theorem is known for all proper subgroups of G , for arbitrary bounded-below coefficient spectra.

Set $Y = N_e^G B$. Proper-family isotropy separation gives

$$\begin{array}{ccc} Y^G & \xrightarrow{\gamma_{G,B}} & F_G(EG_+, Y)^G \\ \lambda_{G,B} \downarrow & & \simeq \downarrow r_{G,B} \\ F_G(E\mathcal{P}_+, Y)^G & \xrightarrow{\gamma_{\mathcal{P},B}} & F_G(EG_+ \wedge E\mathcal{P}_+, Y)^G. \end{array} \quad (4)$$

Here $\gamma_{G,B}$ is the usual comparison, $\lambda_{G,B}$ is induced by $E\mathcal{P}_+ \rightarrow S^0$, $\gamma_{\mathcal{P},B}$ is the comparison after mapping out of $E\mathcal{P}_+$, and $r_{G,B}$ is induced by the projection $EG \times E\mathcal{P} \rightarrow EG$. The fiber of $\lambda_{G,B}$ is

$$F_G(\tilde{E}\mathcal{P}, Y)^G \simeq F_G(S^{\infty \bar{\rho}_G}, N_e^G B)^G = Q_G(B).$$

By [proposition 7.2](#), this fiber is mod- p acyclic, so the left vertical map is a mod- p equivalence.

For a proper subgroup $K < G$, the comparison appearing on a G/K -cell of the lower horizontal map is

$$Y^K \longrightarrow (\mathrm{Res}_K^G Y)^{hK}.$$

The norm restriction formula (1) identifies it with

$$\gamma_{K, B^{\wedge K \setminus G}}: (N_e^K(B^{\wedge K \setminus G}))^K \longrightarrow ((B^{\wedge K \setminus G})^{\wedge K})^{hK}.$$

The coefficient $B^{\wedge K \setminus G}$ is bounded below, so this map is a mod- p equivalence by the inductive hypothesis. Since S/p is dualizable, smashing with it commutes with the homotopy limit obtained by mapping out of $E\mathcal{P}$. Thus the lower horizontal map $\gamma_{\mathcal{P}, B}$ in (4) is a mod- p equivalence.

The right vertical map is an equivalence. Three sides of (4) are therefore mod- p equivalences, so the remaining side $\gamma_{G, B}$ is as well. $\square$

9. THE PROPER TATE DIAGONAL

Proof of theorem 1.2. Let $Y = N_e^G B$, and let

$$u: Y \longrightarrow \beta_G(\iota^* Y)$$

be the Borelification unit, where $\iota^* Y$ denotes the underlying Borel G -spectrum. For every $H \leq G$, the map on categorical fixed points is

$$u^H: Y^H \longrightarrow (\mathrm{Res}_H^G \iota^* Y)^{hH}.$$

For $H \neq e$, (1) identifies this map with

$$\gamma_{H, B^{\wedge H \setminus G}}.$$

Thus theorem 1.1, applied to H and the bounded-below coefficient $B^{\wedge H \setminus G}$, shows that u^H is a mod- p equivalence. For $H = e$, the space EG is nonequivariantly contractible, so the underlying map u^e is the identity under the canonical equivalence and is an equivalence without invoking theorem 1.1.

Categorical fixed points commute with smashing by the finite spectrum S/p . Hence all categorical fixed points of $u \wedge S/p$ are equivalences. The genuine equivariant Whitehead theorem implies that

$$u \wedge S/p$$

is an equivalence in Sp^G ; see [4, Chapter I]. Concretely, the hypothesis says that $u \wedge S/p$ induces an isomorphism on every genuine homotopy group π_k^H , and these groups detect equivalences of genuine G -spectra. Applying the strong symmetric monoidal geometric fixed-point functor Φ^G , we conclude that

$$\Phi^G(u): \Phi^G Y \longrightarrow \Phi^G \beta_G(\iota^* Y)$$

is a mod- p equivalence.

By [8, Example 2.28], this map is precisely

$$\Delta_G(B): B \simeq \Phi^G N_e^G B \longrightarrow (B^{\wedge G})^{\tau G}.$$

Immediately before [7, Lemma 3.15], Quigley–Shah recall the canonical lax symmetric monoidal natural transformation

$$(-)^{tG} \longrightarrow (-)^{\tau G}.$$

That lemma implies that $(B^{\wedge G})^{\tau G}$ is p -complete because G is a finite p -group and $B^{\wedge G}$ is bounded below. Therefore $\Delta_G(B)$ is an S/p -equivalence with S/p -local target. By the universal property of Bousfield localization, an S/p -equivalence from B to an S/p -local spectrum factors uniquely through $B \rightarrow L_{S/p} B$, and the induced map from $L_{S/p} B$ is an equivalence. Hence

$$(B^{\wedge G})^{\tau G} \simeq L_{S/p} B = B_p^\wedge,$$

and $\Delta_G(B)$ is the localization map. $\square$

Remark 9.1 (The trivial group). The restriction $G \neq 1$ is necessary in the proper-Tate formulation. For $G = 1$, the diagonal is the identity of B , which is not the p -completion map unless B is already p -complete.

APPENDIX A. FAILURE OF GENERAL PROPER-TATE POSTNIKOV CONTINUITY

This appendix records a counterexample to a tempting but false strengthening of [proposition 6.2](#).

Proposition A.1. *There is a connective Borel C_2^2 -spectrum E , with trivial action, such that*

$$E^{\tau C_2^2} \longrightarrow \operatorname{holim}_n (\tau_{\leq n} E)^{\tau C_2^2}$$

is not an equivalence.

Proof. Let $G = C_2^2$, $R = H\mathbf{F}_2$, and

$$A = R^{hG} = F(BG_+, R).$$

Then

$$\pi_* A \cong H^{-*}(BG; \mathbf{F}_2) \cong \mathbf{F}_2[x, y], \quad |x| = |y| = -1.$$

The Euler class of the reduced complex regular representation is

$$e = x^2 y^2 (x + y)^2 \in \pi_{-6} A.$$

The Euler-localization formula for proper Tate constructions [1, Proposition 5.10] gives, for a trivial-action R -module M ,

$$M^{\tau G} \simeq M^{hG}[e^{-1}].$$

Take

$$E = \bigvee_{m \geq 0} \Sigma^{6m} R.$$

Since the suspensions tend to $+\infty$, the canonical map from the wedge to the product is an equivalence:

$$E \simeq \prod_{m \geq 0} \Sigma^{6m} R.$$

Consequently,

$$E^{\tau G} \simeq \left(\prod_{m \geq 0} \Sigma^{6m} A \right) [e^{-1}].$$

On the other hand,

$$\tau_{\leq 6n} E \simeq \bigvee_{m=0}^n \Sigma^{6m} R.$$

The multiples of 6 form a cofinal subtower of the Postnikov tower, so passing from all truncation degrees to $6n$ does not change the homotopy limit. Since each displayed wedge is finite,

$$\operatorname{holim}_n (\tau_{\leq 6n} E)^{\tau G} \simeq \prod_{m \geq 0} \Sigma^{6m} A[e^{-1}].$$

The right-hand side contains the degree-zero element whose m -th component is

$$u_m = e^{-m} x^{12m} \in \pi_{-6m} A[e^{-1}] = \pi_0 \Sigma^{6m} A[e^{-1}].$$

The localization of an A -module at e is the telescope of successive multiplications by e . Hence every homotopy class on the left is represented at one finite telescope stage. In the product

this means that all of its components admit one *uniform* denominator e^N . If the element above came from the left, there would therefore be elements v_m such that

$$\frac{v_m}{e^N} = \frac{x^{12m}}{e^m}.$$

The ring $\mathbf{F}_2[x, y]$ is an integral domain, and the nonzero element e is therefore a non-zero-divisor; in particular there is no e -torsion to obstruct cancellation. For $m > N$, cross-multiplication and cancellation of e^N give

$$e^{m-N}v_m = x^{12m}.$$

Since $e = x^2y^2(x+y)^2$, the left-hand side is divisible by $y^{2(m-N)}$, while the right-hand side is not divisible by y , a contradiction. $\square$

Remark A.2. The counterexample does not contradict [proposition 6.2](#). The latter concerns the obstruction functor Q_G on regular norm inputs and uses the geometric-fixed-point formula (2). It also explains why one cannot prove the main theorem merely by applying a hypothetical Postnikov-continuity theorem for proper Tate constructions.

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